

Amit Singh Rajawat

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INTERESTS: Machine Learning, Deep Learning, Computer Vision

EDUCATION

Bachelor of Technology in Information Technology (Artificial Intelligence & Robotics) Aug 2020 - May 2024
Madhav Institute of Technology & Science, Gwalior (CGPA: 8.65) Gwalior, India

- Courses: DBMS, ML and Optimization, OS, Probability and Random Process, CN, Robotics System and Control, CV, NLP.

TECHNICAL SKILLS

Programming: C/C++ (Proficient), Python(Proficient), JavaScript, HTML/CSS, LaTeX, Shell
Tools and Libraries: Latex, Git, Github, Linux, Docker, Pandas, Numpy, Pytorch, Tensorflow, Scikit-learn, Flask, OpenCV, Gstreamer, Dlstreamer, Computer Vision, Reinforcement Learning, OpenVINO Toolkit, Transformers.

EXPERIENCE

Software Engineer June 2024 - Present
Kloudspot Inc., Bengaluru, India Bengaluru, India

- Designed a **Algorithm for Face Recognition Technology Evaluation (FRTE) 1:1 Verification** to submit in National Institute of Standards and Technology (NIST) - Face Recognition Vendor Test (FRVT). **Achieved a global rank of 277** in 1:1 Face Recognition.
- In algorithm firstly face is **detected in image** and then process through **landmark model** where landmarks are detected and then goes for **recognition with second image** and then generate the **similarity score**.
- Created a Algorithm for Face Recognition Technology Evaluation (FRTE) **1:N Identification** to submit in NIST-FRVT.
- Built various models on different datasets using the **ResNet-200** architecture, focusing on optimizing model for face recognition tasks.

Software Engineering Intern Feb 2024 - May 2024
Kloudspot Inc., Bengaluru, India Bengaluru, India

- Created a **high-performance computer vision inferencing framework** in C++ leveraging **GStreamer**, **OpenVINO**, and **DLStreamer**, capable of managing multiple models efficiently, **9% improvement** in processing speed compared to the Python-based implementation.
- Implemented **gRPC** for effective communication between client and server, along with **GStreamer RTSP Server** for seamless real-time video broadcasting which supported high-quality video transmission with minimal packet loss.
- Innovated **pre-processing and post-processing workflows** that refine the flow of video data and improve the accuracy of model predictions.

Project Intern at Indian Space Research Organisation - ISRO (LOR) Dec 2023 - Jan 2024
Indian Space Research Organisation - ISRO, Bengaluru, India Bengaluru, India

- Worked on developing a **vision-based satellite pose estimation system** using the **HRNet algorithm**, which enhanced landmark detection accuracy for **ISRO's upcoming missions**.The system leveraged HRNet's multi-resolution feature, enabling precise detection of key landmarks on satellite images in challenging conditions, such as varying lighting, occlusions, and satellite orientations.
- Engineered the integration of **deep landmark regression** and **nonlinear pose refinement** techniques to accurately determine satellite orientations, training the landmark regression model on extensive satellite image datasets to capture spatial relationships.

Amazon ML Summer School 2023 Sept 2023 - Oct 2023

- DNN, Dimensional Reduction, Casual Interface, RL, Unsupervised Learning, Probabilistic Graphical Models, SL, Sequential Learning

PUBLICATIONS

- A. Rajawat, A. Manjhi and A. Srivastava, "**Categorization of IRIS Flower using Different Machine Learning Strategies**," 2023 IEEE International Conference On Electrical, Electronics, Communication and Computers (ELEXCOM), Roorkee, India, 2023, pp. 1-6 (Link)
- A. Manjhi, A. Rajawat and A. Srivastava, "**Design and Analysis of Programmed Face Monitoring System**," 2023 IEEE International Conference On Electrical, Electronics, Communication and Computers (ELEXCOM), Roorkee, India, 2023, pp. 1-6 (Link)
- Rajawat, A.S., Srivastava, A. "**Recognition of Parkinson's ailment by using various machine learning procedures**". Curr Psychol (2024).(Link)
- On Heart Disease** (Communicated in Journal).

PROJECTS

Snake Game Using Reinforcement Learning With Neural Network (Link)

- Enhanced gameplay using Q-learning**, improving the agent's performance through trial and error with state-space representation and boolean features.
- Visualized agent performance** and evolving strategies with animated plots for a deeper understanding of decision-making.

Text Classification With TensorFlow (Link)

- Developed a high-performance text classification model** for sentiment analysis, and document categorization, leveraging advanced techniques such as NLP and deep learning algorithms. Optimized model performance through hyperparameter tuning, feature engineering, and cross-validation.
- Deployed the optimized model using TensorFlow**, demonstrating advanced machine learning skills in text-based applications.

Rainfall Prediction Using ML (Link)

- Designed a neural network model to predict rainfall** with 95% accuracy, improving predictions through feature engineering and model optimization.
- Provided actionable insights** to support agricultural planning, water resource management, and disaster preparedness.

Movie Recommendation System Using ML With GUI (Link)

- Developed a personalized recommendation system** using collaborative filtering and adaptive logic based on user preferences, trends, and collaborative behavior, with genre-based filtering and weighted ratings for more accurate and relevant recommendations tailored to each user's sear.
- Deployed on a Flask-based GUI** for real-time movie search and personalized recommendations, enhancing user experience with different features.

Hand Gesture Recognition using OpenCV (Link)

- Trained a Temporal Convolutional Network (TCN)** on 12,000 RGB videos to learn efficient temporal modeling in image sequences for recognizing hand gestures, enabling downstream music control tasks such as pause, play, and next.Integration into a real-time application pipeline.
- Achieved 92% accuracy** by pre-processing images with background removal techniques, using color-based segmentation and SIFT for feature extraction

ACHIEVEMENTS

- Leetcode:** 750+ Problem solved & 1765+ Rating (**Profile Link**) , **GFG:** 320+ Problem Solved (**Profile Link**), Featured in **Patrika Newspaper** (Web Link) (Paper Cutting)
- Selected for **IIT Madras Pravartak Undergraduate Fellowship 2023-24 (among 72 in India)** (Link)